WIPRO CAPSTONE PROJECT

NexCare

Customer Service Operations Dashboard

DEVELOPER	STACK	FRONTEND	BACKEND
Rishitha Manyam	Java Full Stack	Angular 17	Spring Boot 3.2

Installation & Run Guide

Step-by-step setup to run the full stack application locally

github.com/rishithamanyam/Capstone-Project

1 Prerequisites

Before running the project, ensure the following tools are installed on your machine:

☕ Java Development Kit

`JDK 17 or higher`

Required for Spring Boot backend. Download from adoptium.net or oracle.com

🪶 Apache Maven

`3.6.0 or higher`

Used to build and run the Spring Boot application

🟢 Node.js

`18.x or higher (LTS)`

Required for Angular development server and npm packages

🅰️ Angular CLI

`17.x (auto-used via npx)`

No global install needed — project uses npx ng commands

Verify your installations by running these commands in a terminal:

```
# Check Java version (must be 17+)
java --version

# Check Maven version
mvn --version

# Check Node.js version
node --version

# Check npm version
npm --version
```

2 Get the Code

Clone the repository from GitHub:

```
# Clone the repository
git clone https://github.com/rishithamanyam/Capstone-Project.git

# Navigate into the project folder
cd Capstone-Project/SIMS
```

📁 The project root is inside the `SIMS/` folder. This is where the Spring Boot application lives. The Angular frontend is in `SIMS/frontend/`.

Project structure overview:

```
SIMS/
├── src/                          # Spring Boot source code
│   └── main/java/com/wipro/csd/
│       ├── config/               # SecurityConfig, DataInitializer
│       ├── controller/           # REST API controllers
│       ├── model/                # JPA entities (User, Ticket, Outage…)
│       ├── repository/           # Spring Data JPA repositories
│       └── service/              # Business logic
├── src/main/resources/
│   └── application.properties  # DB, JWT, port config
├── frontend/                     # Angular 17 app
│   ├── src/app/
│   │   ├── pages/                # login, admin, manager, rep, customer, profile
│   │   ├── shared/               # sidebar, topbar, chatbot, toast
│   │   ├── services/             # auth, ticket, analytics, outage…
│   │   └── guards/               # auth guard (JWT + role check)
│   ├── proxy.conf.json           # Proxies /api → port 8080
│   └── angular.json
└── pom.xml                       # Maven dependencies
```

3 Run the Backend (Spring Boot)

Open a terminal, navigate to the `SIMS/` folder, and run:

A Start the Spring Boot server

```
# From the SIMS/ directory
mvn spring-boot:run
```

Maven will download dependencies on the first run (takes 1–2 minutes). After that, you will see output like:

```
Started CsdApplication in 3.4 seconds (JVM running for 4.1)
```

✅ The backend is now running on **http://localhost:8080**. The H2 database is auto-created and seeded with demo data (users, tickets, outages) on first startup.

⚠️ Keep this terminal open. Do not close it — the backend must stay running while you use the app.

What happens on startup:

- H2 in-memory/file database initialised automatically
- Demo users, tickets and outages seeded via `DataInitializer.java`
- JWT secret key loaded from `application.properties`
- All REST endpoints registered under `/api/**`

4 Install Frontend Dependencies

Open a **second terminal** (keep the first one running the backend) and run:

```
# Navigate to the Angular frontend folder
cd SIMS/frontend

# Install all npm packages
npm install
```

ℹ️ This only needs to be done **once**. After the first install, you can skip this step on future runs.

5 Run the Frontend (Angular)

In the same second terminal (inside `SIMS/frontend/`), run:

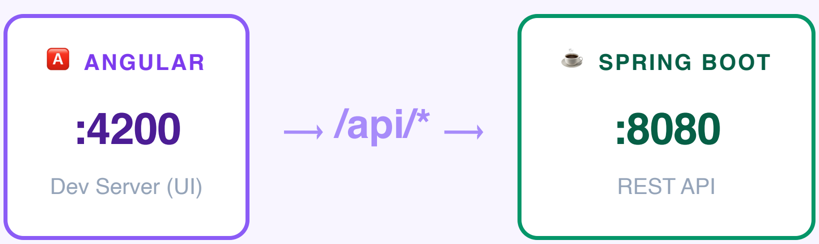
```
# Start Angular dev server on port 4200
npx ng serve --port 4200
```

Wait until you see:

```
✔ Compiled successfully.
Watch mode enabled. Watching for file changes...
```

✅ The Angular app is now live. Open your browser and go to **http://localhost:4200**

How the two servers talk to each other:



`proxy.conf.json` automatically forwards all `/api` requests from port 4200 to port 8080 — no CORS issues, no extra configuration needed.

6 Login & Demo Credentials

Open `http://localhost:4200` in your browser. You will see the NexCare landing page with Sign In and Sign Up options.

Click **Sign In** and use any of the following demo accounts:

ROLE	EMAIL	PASSWORD	REDIRECTS TO
ADMIN	`admin@csd.com`	`admin123`	Admin Dashboard
MANAGER	`john.smith@csd.com`	`Welcome@123`	Manager Dashboard
REPRESENTATIVE	`mike.rep@csd.com`	`Welcome@123`	Rep Dashboard
CUSTOMER	`alice@gmail.com`	`Customer@123`	Customer Dashboard

💡 Each role sees a different dashboard. You can log out using the avatar at the bottom of the sidebar and switch between accounts to explore all views.

7 What Each Role Can Do

A Admin — Full System Access

- View analytics: total customers, managers, reps, tickets, avg satisfaction
- Bar chart: tickets per city | Doughnut chart: tickets per domain
- Add / edit / delete Managers and Representatives
- View all tickets across the system
- Report service outages and mark them as resolved

B Manager — Team Oversight

- View team performance stats (open, in-progress, closed tickets)
- See all tickets assigned to their team's representatives
- Monitor active outages in their area

C Representative — Ticket Handling

- View personal performance stats (response time, resolution time)
- Update assigned ticket statuses (Open → In Progress → Closed)
- Search customers by name, phone number, or customer ID

D Customer — Self-Service Portal

- View own ticket stats and history
- Raise new support tickets (domain → subject → description)
- See available service plans (₹499 / ₹799 / ₹1199 per month)
- Receive outage alerts if there is a disruption in their city
- Rate resolved tickets with a 1–5 star rating
- Chat with NexCare virtual assistant (chatbot) for instant support
- Register a new account via Sign Up on the login page

8 Common Issues & Fixes

PROBLEM	LIKELY CAUSE	FIX
Backend won't start	Java version below 17	Install JDK 17 or higher from adoptium.net
Port 8080 already in use	Another app on 8080	Kill the process or change port in `application.properties`
npm install fails	Node version too old	Upgrade Node.js to v18 LTS or higher
API calls return 404	Backend not running	Check Terminal 1 — Spring Boot must be running on :8080
Login fails with valid credentials	DB not seeded (already has data)	Delete the H2 `.db` file in project root and restart backend
White screen on load	Angular compile error	Check Terminal 2 for ng serve errors. Run `npx ng build` for details

9 Quick Reference

ITEM	VALUE
App URL	`http://localhost:4200`
API Base URL	`http://localhost:8080/api`
GitHub Repo	`github.com/rishithamanyam/Capstone-Project`
Backend command	`mvn spring-boot:run`
Frontend command	`npx ng serve --port 4200`
Backend port	`8080`
Frontend port	`4200`
Database	H2 (auto-created on startup)
Auth type	JWT (stored in sessionStorage)

🎯 **NexCare — Customer Service Operations Dashboard**

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